

UNIVERSITY OF MIAMI

SPEECH TRANSMISSION INDEX ESTIMATION FOR NON-CLASSROOM
SPACES

By

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Speech Transmission Index Estimation
for Non-Classroom Spaces

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The Speech Transmission Index (STI) is a useful acoustic parameter of a space that gives a metric of how well speech is understood. Having access to a way to estimate an acoustic parameter is often highly desired, just as Sabine did with his famous reverberation equation. Human speech is a function of several forms of modulation: pitch, amplitude, and temporal modulation. The preservation of these modulations in a space is known as the Speech Transmission Index. Having a way to predict parameters is highly desired; much like Sabine's famous RT60 equation, the STI has its own existing prediction equations.

The existing STI equations were derived by performing a series of classroom measurements, most of which were performed in primary school classrooms. This resulted in several equations that yielded high prediction accuracy, specifically in classroom conditions. Although classrooms are one of the best examples of where STI should be predicted, they are far from the only spaces that benefit from STI predictions.

The following paper investigates the accuracy of the existing STI equations in concert halls, in order to stress test them in a different acoustic environment from a standard classroom.

The first part of this thesis investigates the mathematics behind the development of STI and its calculation, other useful speech-related acoustic parameters, and a

thorough dive into the type of spaces used to develop each STI equation. It also explains the motivation behind this work and the current methods for measuring STI.

The second part of this thesis describes the methods used to calculate and measure the STI in both pre-recorded and physical spaces. The tools used to capture STI data, the statistical methods used to analyze it, and the reason for the use of multiple linear and Gaussian process regression models for the development of the new models.

The third part of this thesis explores the results from the new data and the accuracy of the new proposed models, the performance of the models calculated using the root mean squared error and mean absolute error.

In conclusion, this thesis presents the accuracy of the existing equations in larger and more reflective spaces. It provides a new dataset of STI and frequency band measurements of 16 different venues, new models that can be further improved upon to increase their accuracy in this acoustic environment, and discusses where future work in this field should go.

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List of Abbreviations

ANOVA	Analysis of Variance
C50	Clarity Index
CSV	Comma-Separated Values
D50	Definition (Deutlichkeit) Index
dB	Decibels
G.R.A.S.	Gunnar Rasmussen Acoustic Systems
GPR	Gaussian Process Regression
IR	Impulse Response
MAE	Mean Absolute Error
MLP	Multi-Layer Perceptron
MLR	Multiple Linear Regression
REW	Room EQ Wizard
RIR	Room Impulse Response
RMSE	Root Mean Squared Error
RT	Reverberation Time
RT60	Time it takes for a sound to drop by 60 dB
SII	Speech Intelligibility Index
SNR	Signal-to-Noise Ratio
SPL	Sound Pressure Level
STI	Speech Transmission Index
STIPA	Speech Transmission Index of P.A. Systems
T20	Time it takes for a sound to drop by 20 dB
T30	Time it takes for a sound to drop by 30 dB

CHAPTER 1

Introduction

1.1 Motivation

As of now, the most accurate way to measure speech transmission index (STI) is to go into an existing room and use a meter to measure the STI according to some International Organization for Standardization (ISO) standards [1]. While there is software, such as ODEON and EASE [2, 3] that can predict the STI of a yet to be constructed room, it is enormously expensive, and still requires large amounts of time to model the space to get the prediction. Having a quick and easy equation to determine the STI could be a valuable tool, allowing acousticians the ability to assure a client that the speech intelligibility could be greatly improved if certain materials are used or changed. The earlier an acoustician knows this, the easier it would be to make informed edits to architectural documents to improve the acoustic and aesthetic experience of those who use the space.

A few equations have been developed to try and estimate the STI of classrooms, the typical use case for good STI [4–7]. However, there are many other spaces that require careful calibration of STI as well. The researchers that developed these equations measured many primary school classrooms, some lecture halls, and a very small

amount of large more reverberant spaces. This allowed the equations to have high predictive accuracy in the more acoustically controlled classroom environment. Yet there are a plethora of other spaces that indeed need careful calibration of STI, such as restaurants, food courts, airports, and meeting rooms. This thesis investigates the robustness of the existing equations in large reflective spaces, those being concert halls at the University of Miami. Much like Sabine’s famous reverberation equation, having a quick and easy way of determining the STI of a space allows for a more streamlined development stage, especially for the spaces where STI is often problematic, like food courts or airports.

1.2 Background

In the late 1890s, Wallace Clement Sabine developed his famous reverberation equation, which allowed for the estimation of the reverberation time of a room. The equation states that the decay of sound by 60 decibels, known as the RT60, is proportional to the total absorption of the room and the volume of the room [8].

$$\text{RT60} = 0.049 \cdot V/A \quad (1.1)$$

Where the total absorption is given by:

$$A = \sum_i S_i \cdot \alpha_i \quad (1.2)$$

Using this equation, Sabine crafted Boston’s “Symphony Hall” to have one of the most exquisite sounds for an orchestra [8], thus Symphony Hall is regarded as one of the best concert halls in the world, along with Vienna’s Musikverein and Amsterdam’s

Concertgebouw. Despite later discoveries about its drawbacks, the predictive ability of Sabine’s equation remains significant.

Since then, other equations and models have been produced that can estimate the RT60 of a room with greater accuracy; however, even a simple equation like his can lead to monumental achievements in acoustic design [8, 9]. Therefore, having access to a quick and simple estimation for various parameters in a space is highly desired. This paper provides the background of another metric for assessing acoustic spaces, the Speech Transmission Index, and investigates the accuracy of existing STI estimation equations in a variety of spaces.

It is important to understand how and why the Speech Transmission Index works, its origins, and the evolution of its measurement methods. STI is a measure indicating how well human speech is understood in a space, with values between 0 and 1, where 0 implies unintelligibility and 1 implies perfect intelligibility [10]. This scale is traditionally divided into five categories, where an STI between 0 and 0.3 is considered “Bad”, between 0.3 and 0.45 is “Poor”, between 0.45 and 0.6 is “Fair”, between 0.6 and 0.75 is “Good”, and anything above 0.75 is “Excellent”.



Figure 1.1: STI scale.

The original developers of STI were Tammo Houtgast and Herman Steekneken; they were tasked to perform range measurements on radio systems. The task was a tedious one, so they developed a method to predict the intelligibility of speech over these radio systems to make this monumentally large and arduous task easier, and thus STI was born. The specifics of how it is calculated can be seen in their

textbook *Past, Present, and Future of STI* [10] and will be discussed later. Like all methods and technologies, adaptations to STI have been made since then to improve its predictive power; various other methods of calculating speech intelligibility have also been developed such as Room Acoustic STI (RASTI), Speech Intelligibility Index (SII), and even Neural Network models [11, 12]. A few other important metrics that relate to speech and intelligibility are the Clarity Index (C50) and the Definition Index (D50).

1.2.1 RASTI

The Room Acoustic Speech Transmission Index (RASTI) is a condensed version of the original STI. It was developed to take less time to measure by using only two octave bands, 500 Hz and 2000 Hz, instead of the full seven octave bands traditionally used in STI measurements. For this reason, RASTI is less frequently utilized compared to traditional STI. Additionally, deviations between STI and RASTI occur most significantly under conditions with low speech-to-noise ratios [13]. Therefore, in rooms susceptible to higher noise levels, it is preferable to rely on full STI measurements.

1.2.2 SII

The STI is built off of the Modulation Transfer Function (MTF), which is described in detail in a later section. This MTF has a weighted average applied to it to produce the STI, this average derived by [14], where psychoacoustic tests were used to find which octave bands affected the perception of speech the most. STI therefore is more about the perception of speech to humans than the objective measure. An

analogy could be made here between pitch and frequency, where low or high pitches can be perceived differently depending on the volume they are played at. While pitch can perceptibly be changed, frequency does not, as it is an objective measure. In this analogy SII is like frequency, an objective measure of the overall intelligibility of speech; STI is like pitch, a subjective, perceptual measure of how the typical human hearing system would hear speech in a space or over a communications system.

1.2.3 STIPA

If a room is large, like a lecture hall, and has a Public Address system (PA) the standard STIPA test signal, the warble tone as it is often called, could be played over the PA system in that space, and an STI meter would listen for this amplified test signal to calculate the STIPA (Speech Transmission Index for Public Address systems). Like the RASTI, STIPA is a condensed version of the full STI. While STIPA indeed uses all seven octave bands in human speech, it only uses 2 modulation frequencies rather than the full 14 used in full STI.

1.2.4 STI

The speech transmission and intelligibility issues that Houtgast and Steekneken faced with the radio systems have their own direct parallels to acoustic spaces; for instance, phase modulations that could occur over a telecom system relate to how a room's reverberation can get in the way of spoken word in a highly reverberant space. Likewise, any background noise from a radio signal can interfere with the clarity of the voice over the radio, just as background noise interferes with a listener's ability to hear a lecturer in a classroom. Thus, the STI methods developed by Houtgast

and Steekneken have their direct correlation to room-acoustic STI calculations as well. With the advancement of technology, meters were soon developed to be able to measure a room's STI.

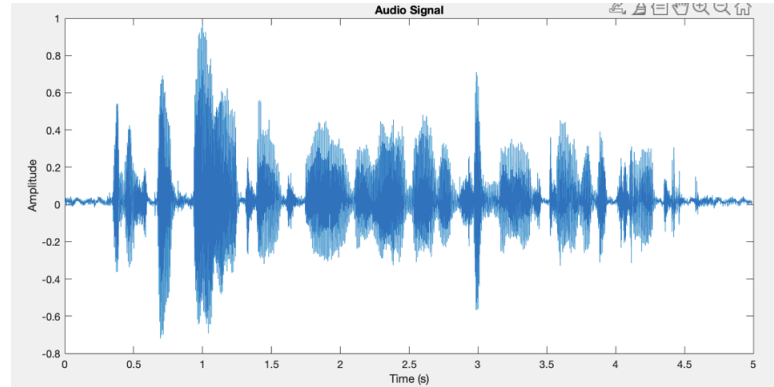


Figure 1.2: Speech Signal.

As shown in the figure above, speech is made up of both amplitude, temporal, and frequency modulation. The phonemes spoken are displayed on the horizontal axis (temporal); these phonemes, indicated by the spikes of blue, have pitch (frequency) and amplitude, which can be seen on the vertical axis of this figure. At any given moment, the sounds we make to speak are composed of the modulations of these three features; thus, speech is a modulating signal [15]. These modulations are crucial to speech signals; the preservation of these modulations is what ensures the speech is understood on the other end of the telephone, or at the last row of seats in a lecture hall. Meters such as the NTI meter, specifically “listen” to a test signal, often called the warble tone or the STIPA signal, which is a modulating signal that mimics the frequency and amplitude modulation of speech, to determine how well that room or telecom system preserves speech modulations. The result from this calculation will give that position in the space an STI ranking between 0 and 1. To get an accurate

measurement, it is best to follow the ISO standards found in [1]. In this manner, one could take an average of all the measurements in a room to get the STI of that space.

It is important to understand fundamentally how STI is calculated, specifically how it is calculated in an existing space. The calculation is defined in [10, 16–18], and will be implemented in MATLAB code in detail in the methods section of this paper. As described above, speech and its intelligibility has much to do with how well preserved the modulations of speech travel “from the speaker’s mouth to the listeners’ ears...” [10]. Equation 1.3 displays the official STI equation developed by Steekneken and Houtgast.

$$STI = \sum_{k=1}^7 (W_k \cdot TI_k) \quad (1.3)$$

STI is the sum of the product of modulation weights and the transmission indices for the 7 octave bands in the human speech spectrum. The transmission indices for the 7 octave bands are:

$$125\text{Hz}, 250\text{Hz}, 500\text{Hz}, 1000\text{Hz}, 2000\text{Hz}, 4000\text{Hz}, 8000\text{Hz} \quad (1.4)$$

The Weight W_k is the octave band weighting factor for the given octave band k , and the Transmission Index, TI_k , indicates the modulation depth preservation in proportion to the signal to noise ratio (SNR) for the given octave band k . The octave band weighting factors were determined by French and Steinberg in [14]; they are as follows: $W_1 = 0.13$, $W_2 = 0.14$, $W_3 = 0.11$, $W_4 = 0.12$, $W_5 = 0.19$, $W_6 = 0.17$, $W_7 = 0.14$. As described earlier, these weights are used to accommodate for psychoacoustic artifacts that the human hearing system imprints on the incoming speech signal. By converting a room impulse response (RIR) into the frequency domain using the Fourier transform, normalizing it, filtering the speech signals using each of the seven

octave bands, and applying the Modulation Transfer Function to it according to the known weighting factors W_k , a room acoustic STI calculation can be determined.

1.2.5 Clarity Index and Deutlichkeit (Definition) Index

The Clarity Index or C50, is a measure of the clarity of speech. When the reverberation of a space increases this leads to more unintelligible speech, however under certain conditions the reverberation can potentially be beneficial to the speech or at a minimum can be negligible to the intelligibility of speech. If the delayed speech does not exceed 50 ms, the delay can actually act positively on the intelligibility of the speech. There are a few room acoustic conditions that can help the C50, these conditions are: a rear wall that can absorb the late reflections of the speech, narrow walls that are close to the speaker as they speak, a room that maintains a short distance between the speaker and the listeners, and a low signal to noise ratio. A higher C50 indicates that there is more early energy than late energy, thus resulting in better speech intelligibility.

In [19], it was found that a change in 1.1 dB is the just noticeable difference of C50, this results in an average change in 0.03 STI units. As mentioned earlier, distance has a significant effect on the C50 index of a space; therefore C50 is often coupled with distance measurements to reveal issues in large venues such as concert halls. The Deutlichkeit Index, often referred to as the Definition Index or D50, is related to C50 in that it can be used to calculate the C50 given the following equation:

$$C_{50} = 10 \log_{10} \left(\frac{D_{50}}{1 - D_{50}} \right) * 100\% \quad (1.5)$$

Where the D50 is defined by:

$$D_{50} = \frac{E_{\text{early}}}{E_{\text{total}}} \quad (1.6)$$

This states that the D50 is a ratio of the early energy over the total energy. A higher D50 results in better clarity which in turn results in better speech intelligibility.

1.2.6 Establishing The Knowledge Gap

In [4], researchers Leccese, Rocca, and Salvadori investigated the accuracy of five different equations, one of which they developed themselves as a part of their experimental process. Equation 4.5 found in [4], was originally developed by Tang, Shiu Keung, and M. H. Yeung in [6], below is their developed equation:

$$STI = 0.5895 - 0.4422 \log_{10} (RT60_{500 \text{ Hz}}) \quad (1.7)$$

Equations 4.6 and 4.7 found in [4], are respectively linear and logarithmic equations developed by Escobar and Morillas [7].

$$STI = 0.778 - 0.143 (RT60_{500 \text{ Hz}}) \quad (1.8)$$

$$STI = 0.634 - 0.192 \ln (RT60_{500 \text{ Hz}}) \quad (1.9)$$

Equation 4.8 found in [4], was developed by Nowoswait and Olechwska as seen in [5].

$$STI = 0.6488 - 0.2078 \ln (RT60_{500 \text{ Hz}}) \quad (1.10)$$

Lastly, Leccese, Rocca, and Salvadori, developed what they called the “new logarithmic equation” 4.9, found in [4].

$$STI = 0.5940 - 0.1827 \ln (RT60_{500 \text{ Hz}}) \quad (1.11)$$

Other than one exception, which is equation 4.6 found in [5], the papers that cover the subject of predicting STI, base their measurements primarily on classrooms or educational spaces [4–7]. For instance, in [6] Tang et al. measured classrooms in 18 different primary schools, in [4] Leccese et al. measured 11 classrooms to verify the accuracy of equations derived in [5–7], and in [7] another 17 educational rooms of varying sizes were measured. Some of the equations were derived solely from data gathered in primary school classrooms, which are typically smaller spaces with fewer reflective materials. Other papers take size into account and include large lecture halls in their measurements [4, 5, 7]. However, of the four papers where the five equations above came from, only one paper included non-classroom spaces to develop their STI estimation equation [5]. This paper had its own limitations, the most significant being that it measured only six spaces in total (four classrooms, one warehouse, and one church). The details explained here are not to take away from the work that each researcher has done, their methods are sound and produced accurate results, as shown in [4]. However, if all the data that is collected for this STI estimation is based in these lower reverberation spaces that are typically small, then it would be reasonable to conclude that the accuracy of the equations would drop in a more extreme acoustic environment.

An argument could be made: Knowing the STI in larger more reflective spaces can be a valuable thing, for instance, churches, airports, restaurants, and even concert venues. This metric can be beneficial in ensuring that speech is not too distracting for people sitting at a restaurant, or guests at an airport being able to hear zone numbers being called out.

While in many of these cases, amplification might be the solution to the speech transmission problem, having a metric beforehand can help influence how much amplification is needed, and can help guide acousticians to make choices for the size and materials of the space if amplification is not what the client desires due to aesthetic reasons. As mentioned before, a space such as a restaurant typically would aim to lower the speech intelligibility, thus in this case being able to lower the predicted STI value would be ideal. Such a case would most likely be the primary use of this equation, where acousticians know that a space is highly reverberant, with poor STI, and by lowering the reverberation would lead to an improved STI. Papers such as [8, 9] reveal that room geometry influences the reverberation of the room.

In equation 4.1, it can be seen that as the volume of a space increases, the RT60 likewise will increase; intuitively this is known because volume has so much influence over the reverberation of a space. [9] takes this concept a step further by showing that Sabine's equation is not perfect and does not take into consideration the room geometry. [9] shows that room geometry can affect the RT60 even when the volume is a constant metric. To explain this detail further, if two spaces had equivalent room volume, but one space has statues, and balconies, and other objects in the room that act as diffusers of sound, while the other space is an empty room, it becomes easy to understand where the error in Sabine's equation lies. If this is the case, and the existing STI equations are a function of the reverberation time at a specific frequency band, then it is logical to assume that the speech transmission index would be different in these much larger spaces. Thus, the equations potentially could be adapted to include these other spaces and increase their predictive power.

CHAPTER 2

Methods

For this thesis there were two stages of measurement; one was a simulation phase using the Reverbdata dataset, the other was a measurement phase performed on campus at the University of Miami.

2.1 Simulation Phase

Twelve total directories were created, each to store the RIR data for each venue in the Reverbdata dataset. Three MATLAB scripts were used to gather, process, and analyze the wav files in each of these directories. First a script was written that could implement the STI calculation as discussed in the background section of this thesis. This script iterated through all the wav files in the directory of the desired hall or location, calculating an STI value for each wav file. Each wav file represented a different position in the venues. The method used to calculate the STI follows the exact protocol that can be found in these sources [10,16,18,20]. The script implements a function, which creates variables of octave band center frequencies starting from 125 Hz up to 8 kHz. Next using MATLAB's built-in octave filter, these center frequencies

are used alongside the modulation frequencies determined by the following equation:

$$\text{Modulation Frequencies} = 2^{\left(\frac{n}{3}\right)}, \quad n \in \{-2, -1, 0, \dots, 11\} \quad (2.1)$$

to compute the Modulation Transfer Function (MTF) for each of these center frequencies. In turn this creates a filtered room impulse response for each octave band; each of these filtered MTFs is stored into a matrix. The most important frequencies for human speech exist between 1.5 kHz and 4 kHz, thus a weighted average was applied to the MTFs in order to compute the STI. An array was created with these weights as shown in [14]; these weights were then multiplied with the MTFs to finally calculate the STI. The individual STI values were saved into a CSV file within the directory containing that room's specific RIR wav files. This method was congruent with how the real measurements were also taken in the four concert halls on campus.

The second script likewise also iterated through the same directories of venues in the Reverbdata dataset. Since REW does not have a built-in function to average RT data across many measurements, this MATLAB script was developed to perform this task. The script too iterated through the same directory of wav files and calculated the RT60 values at 500 Hz for each file. Similarly to the previous script, this script also used MATLAB's built in octave filter function to filter the RIR, however this one only used 500 Hz as the center frequency. The script then implemented the Schroeder curve method to calculate the RT60.

It started by normalizing the impulse responses, converting the magnitudes to decibels, then used the Schroeder Curve method. This method takes the computed decibel levels, and reads them in reverse order, thus starting with the noise floor of the room as a reference value. The script uses T30 and T20 to compute the RT60,

where

$$RT60 = T20 \times 3 \quad (2.2)$$

$$RT60 = T30 \times 2 \quad (2.3)$$

The T20s and T30s are computed by searching the computed levels to find when in the impulse response recording was the level -5 dB_{SPL}, and -25 dB_{SPL} (for T20) or -35 dB_{SPL} (for T30). The difference in time between the -5 and the -25 or -35 dB_{SPL} determines the T20 or the T30 respectively. T30 is generally considered to be more accurate due to more of the signal decay being used in the calculation, therefore if a T30 was found, this would be the value used to calculate RT60. If only a T20 can be found then the T20 would be used to calculate the RT60. The results were saved into a separate CSV file for further analysis.

The final script read the RT60 values from the generated RT60 at 500 Hz CSV, created by the previous RT60 script, and used the values as input for a function that would implement all five of the existing STI estimation equations found in [4–7]. Thus resulting in all five equation’s predicted STI values for each position in each venue of the Reverbdata dataset. An average of all the STI predictions was also used given by the following equation:

$$STI_{AVG} = (STI_{1.3} + STI_{1.4} + STI_{1.5} + STI_{1.6} + STI_{1.7} + STI_{1.8})/5 \quad (2.4)$$

Once all of the CSV files were created, they were opened in Microsoft’s Excel. Each venue had a chart, which was a side by side comparison between the calculated STI and the predicted STI for each wav file. Using these charts, paired *t*-tests and *t*-tests for correlation were performed on each respective hall’s calculated and predicted STIs. The paired *t*-tests were used to find if there was a significant difference between

calculated and predicted STI values. A large gap between the calculated and predicted STI values would result in a p -value that was below the α threshold of $\alpha = 0.05$. Correlations were then calculated to give an r value that could be passed into a t -test for correlation. A p -value from the correlation t -test that was below the α threshold of $\alpha = 0.05$ indicated an STI prediction that had a significant correlation to the calculated STI values. Both of these processes were repeated using all six of the STI equations, the original five STI estimation equations and an average of all five predictions being the sixth equation.

2.2 Measurement Phase

This section will describe how the room measurements in each of the four locations on campus at the University of Miami were gathered.

2.2.1 Calibration

Once the simulation phase was completed, the room measurements were the next step in the experiment. Before performing any room measurements, it was important to first calibrate the entire measurement system to ensure that none of the audio equipment left any artifacts in the measurements. Using REW, a calibration file for the Babyface pro FS audio interface was generated, by plugging in the output of the interface into its own input. The G.R.A.S 26CA microphone calibration file was imported into REW; the reported REW level and the Larsen Davis SPL meter were calibrated with each other. This calibration process, between REW and the microphone, was repeated in each location on campus. The speaker calibration process was performed in an anechoic chamber located at the Motorola Solutions fa-

cility. This chamber was used to measure the frequency response of the Yamaha HS7 Studio Monitor. By performing sine sweeps in the anechoic chamber, the captured sound theoretically has no artifacts imprinted onto it by reflections that traditionally would exist if the sweeps were performed in a regular reflective space. Thus the captured sound is purely the frequency response of the source, that source being the HS7 connected to the entire measurement system.

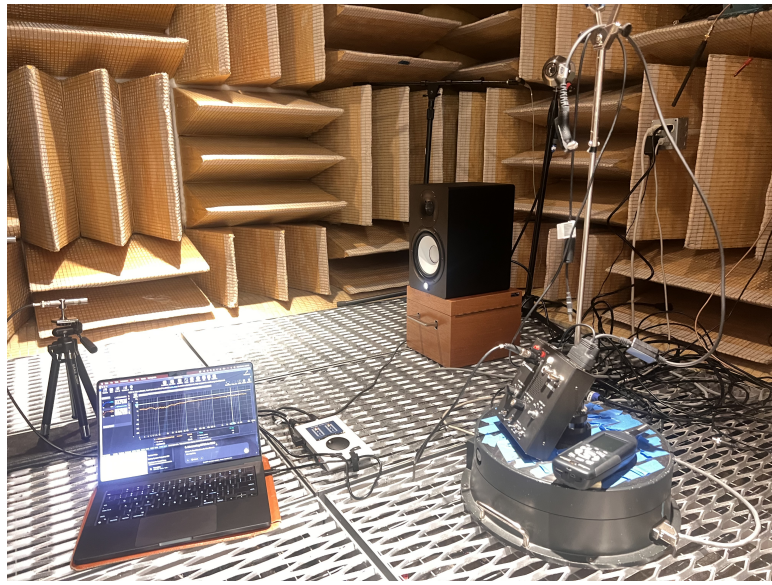


Figure 2.1: Yamaha HS7 Anechoic Chamber Calibration

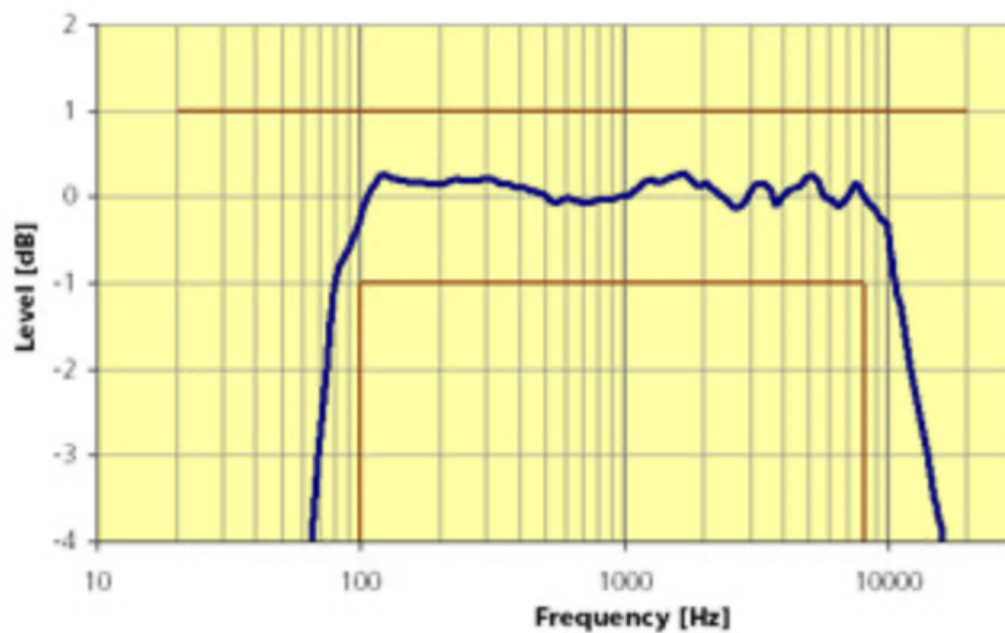


Figure 2.2: NTi Talkbox Frequency Response.

The natural frequency response of the HS7 is shown here:

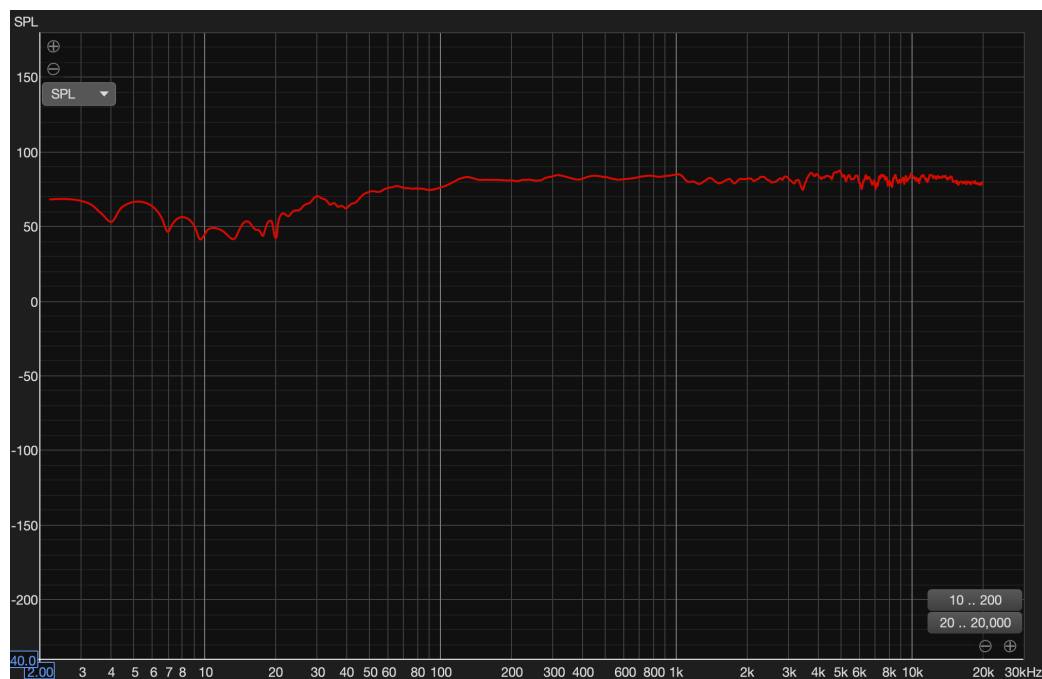


Figure 2.3: Yamaha HS7 Frequency Response

Both the NTi Talkbox and the HS7 naturally had flat frequency responses within the 100 Hz to 10 kHz range. It was important to verify that the HS7 had a similar response to the NTi Talkbox, as a talkbox is often used as the reference signal generator for STI and STIPA measurements.

2.2.2 STI Studio Six Digital Calibration Procedure

There were two separate calibration processes, one for each measurement type. For the STI measurement, the STI/STIPA signal was played out of the speaker; standing one meter from the speaker, the SPL meter was to read 65 dB_{SPL}.

2.2.3 REW Sine Sweep Calibration Procedure

For the RT60 measurements which used a sine sweep, REW has a built-in SPL meter with a calibration process. The measurement microphone too was also placed one meter from the source. Upon hitting the calibrate button, REW would begin to play a signal. The SPL meter too was placed at one meter from the speaker, the output gain was adjusted until the SPL meter read 80 dB_{SPL}, this 80 dB_{SPL} metric was recorded into REW and used as the calibration for the microphone.

2.2.4 Measurement procedures

Taking the HS7, Babyface Pro, G.R.A.S 26CA, into each concert hall on campus the procedure followed this protocol:

- The HS7 was placed in the room/concert hall at the location where a speaker would typically be standing. Each hall had a podium on which the HS7 was placed to mimic the height at which a human mouth would be speaking at.

- The HS7 and the G.R.A.S 26CA were connected to the Babyface pro FS, then the Babyface pro FS was connected to the laptop.
- The speaker, microphone, interface were calibrated for the STI measurement.
- Using the StudioSixDigital STI/STIPA app, the STI signal (warble tone) was played through the HS7, the StudioSixDigital application would then run a measurement.
- Once the four STI measurements at the current position were gathered the speaker and microphone were then calibrated for the RT60 measurements.
- Using REW, four sine sweeps were then recorded at the same position, thus allowing for each STI measurement to have a respective sine sweep.
- This process was repeated at 8 different locations in each hall, resulting in a total of 64 measurements.
- After all the STI and sine sweep measurements were taken in the concert halls, four wood clap RIRs were recorded in each concert hall using the G.R.A.S 26CA, Babyface Pro FS, and Logic Pro.



Figure 2.4: G.R.A.S 26CA capturing STI measurement.



Figure 2.5: G.R.A.S 26CA capturing sine sweep measurement.

To gather the volume measurements for all the halls on the campus, blueprints as well as a laser measuring tool. The blueprints are shown in figures 2.5 and 2.6. Using the laser measurement tool, the distance between the floor and ceiling were measured at important locations such as the front and back of the stage, as well as the first row of seats and the last row of seats. Using these height measurements and EasyPDFTakeOff, a software that allows for the measurement of distances using blueprints, the volume of each concert hall was calculated. No blueprints were given for Clarke Recital Hall, so the volume measurement was manually calculated.

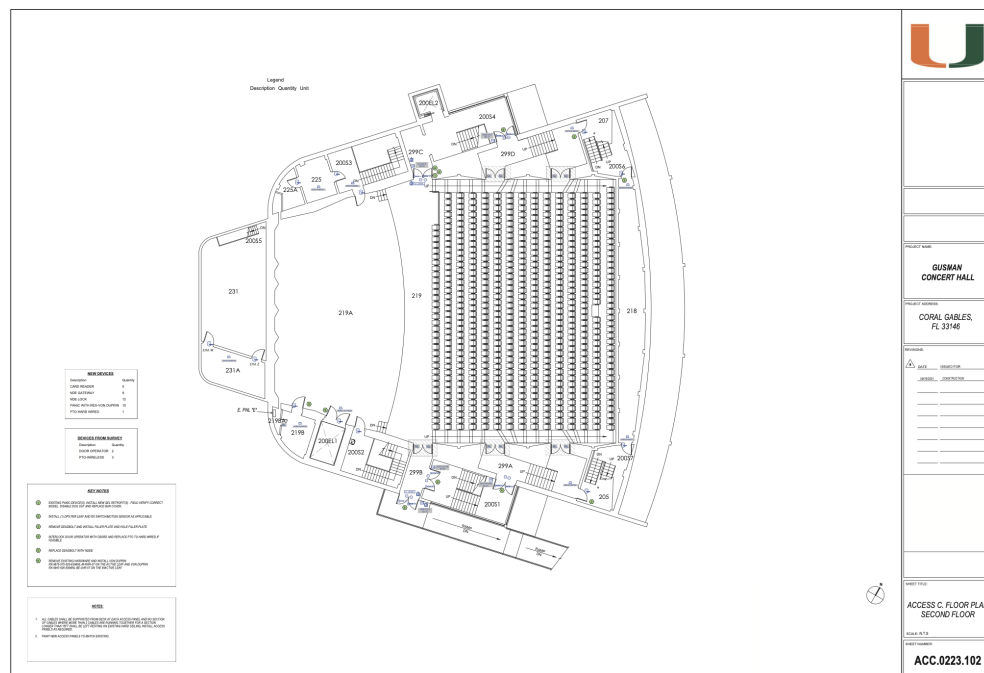


Figure 2.6: Maurice Gusman Hall Blueprint.

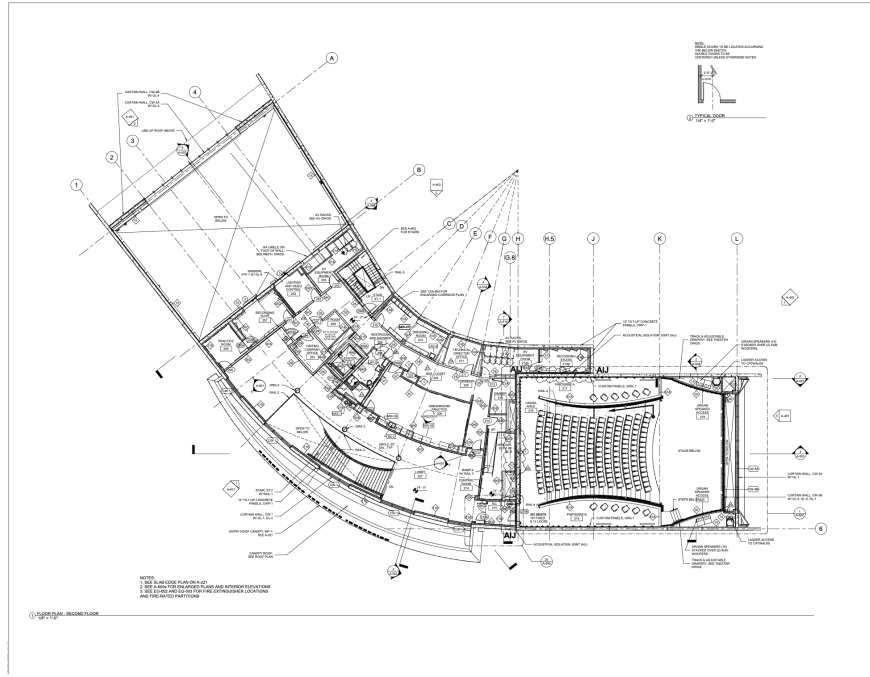


Figure 2.7: Hormel Innovation stage and Newman Recital Hall Blueprints.

2.2.5 Analysis and Model Creation

After analysis of the data measured, a few models were developed to provide a more accurate method of estimating STI for concert hall like spaces. To provide the models with more features the scripts previously mention in the simulation phase section of this thesis were adapted to calculate the RT60s across all seven octave bands in human speech, rather than simply calculating the RT60 at 500 Hz.

Below displays the new original table with all averaged RT60 data for all seven octave bands:

Venue	STI	Seats	Vol. (m ³)	Ac. Dens.	125Hz	250Hz	500Hz	1kHz	2kHz	4kHz	8kHz
T60 (s)											
Cartuxa Monastery Church	0.40	240	4475	0.05	1.66	1.52	1.13	1.03	0.75	0.41	0.19
Coliseu Micaelense	0.48	1300	6000	0.21	2.35	1.23	1.17	1.11	0.88	0.50	0.21
Convent of Christ Church	0.21	–	5500	–	2.61	2.64	2.47	2.05	1.38	0.68	0.29
Convent of Christ Cistern	0.32	–	767	–	4.67	3.15	2.56	1.87	1.22	0.62	0.27
Dom Joao III	0.43	–	5000	–	1.94	1.82	1.65	1.35	0.93	0.47	1.82
ESML Great Hall	0.50	448	4527	0.10	2.90	3.12	2.56	1.66	0.83	0.69	0.42
ESML Organ Room	0.53	–	650	–	0.75	0.80	0.73	0.69	0.63	0.39	0.19
ESML Small Hall	0.64	56	396	0.14	0.51	0.48	0.47	0.45	0.44	0.42	0.31
Gulbenkian Grand Hall	0.61	1203	7000	0.17	0.75	0.80	0.83	0.79	0.70	0.42	0.26
Lourinha Misericordia Church	0.47	50	1850	0.03	0.61	0.74	0.81	0.80	0.68	0.46	0.22
St Dominic Convent Church	0.48	–	6582	–	1.30	1.74	1.62	1.39	0.91	0.48	0.22
St George Cinema	0.53	830	3600	0.02	0.64	0.68	0.65	0.53	0.48	0.29	0.14
Newman	0.62	200	3350	0.06	1.87	1.51	1.68	1.76	1.58	0.41	0.22
Hormel	0.66	300	1710	0.18	1.65	1.31	1.08	1.10	1.16	0.41	0.23
Clarke	0.68	147	1300	0.11	1.03	0.82	0.74	0.75	0.75	0.41	0.22
Gusman	0.39	600	5,500	0.11	2.24	1.77	1.88	2.03	1.90	0.41	0.21

Table 2.1: Original Seven Octave Band Data

The weighting factors determined by [14] were applied to the seven octave band frequencies in the original chart, thus resulting in the chart below:

Venue	STI	Seats	Vol. (m ³)	Ac. Dens.	125Hz	250Hz	500Hz	1kHz	2kHz	4kHz	8kHz
T60 (s)											
Cartuxa Monastery Church	0.40	240	4475	18.65	0.22	0.21	0.12	0.12	0.14	0.07	0.03
Coliseu Micaelense	0.48	1300	6000	4.62	0.31	0.17	0.13	0.13	0.17	0.08	0.03
Convent of Christ Church	0.21	–	5500	–	0.34	0.37	0.27	0.25	0.26	0.11	0.04
Convent of Christ Cistern	0.32	–	767	–	0.61	0.44	0.28	0.22	0.23	0.11	0.04
Dom Joao III	0.43	–	5000	–	0.25	0.26	0.18	0.16	0.18	0.08	0.25
ESML Great Hall	0.50	448	4527	10.10	0.38	0.44	0.28	0.20	0.16	0.12	0.06
ESML Organ Room	0.53	–	650	–	0.10	0.11	0.08	0.08	0.12	0.07	0.03
ESML Small Hall	0.64	56	396	7.07	0.07	0.07	0.05	0.05	0.08	0.07	0.04
Gulbenkian Grand Hall	0.61	1203	7000	5.82	0.10	0.11	0.09	0.09	0.13	0.07	0.04
Lourinha Misericordia Church	0.47	50	1850	37.00	0.08	0.10	0.09	0.10	0.13	0.08	0.03
St Dominic Convent Church	0.48	–	6582	–	0.17	0.24	0.18	0.17	0.17	0.08	0.03
St George Cinema	0.53	830	3600	4.34	0.08	0.09	0.07	0.06	0.09	0.05	0.02
Newman	0.62	200	3350	–	0.24	0.21	0.18	0.21	0.30	0.07	0.03
Hormel	0.66	300	1710	–	0.21	0.18	0.12	0.13	0.22	0.07	0.03
Clarke	0.68	147	1300	–	0.13	0.12	0.08	0.09	0.14	0.07	0.03
Gusman	0.39	600	5,500	–	0.29	0.25	0.21	0.24	0.36	0.07	0.03

Table 2.2: Weighted Seven Octave Band Data

A similar dataset was made that applied the inverse of the weights. Shown below:

Venue	STI	Seats	Vol. (m ³)	Ac. Dens.	125Hz	250Hz	500Hz	1kHz	2kHz	4kHz	8kHz
					T60 (s)						
Cartuxa Monastery Church	0.40	240	4475	18.65	1.45	1.30	1.01	0.91	0.61	0.34	0.16
Coliseu Micaelense	0.48	1300	6000	4.62	2.04	1.06	1.04	0.98	0.71	0.41	0.18
Convent of Christ Church	0.21	–	5500	–	2.27	2.27	2.20	1.80	1.12	0.56	0.25
Convent of Christ Cistern	0.32	–	767	–	4.06	2.71	2.28	1.64	0.99	0.52	0.23
Dom Joao III	0.43	–	5000	–	1.69	1.57	1.47	1.19	0.75	0.39	1.56
ESML Great Hall	0.50	448	4527	10.10	2.52	2.68	2.28	1.46	0.67	0.57	0.36
ESML Organ Room	0.53	–	650	–	0.65	0.69	0.65	0.61	0.51	0.33	0.16
ESML Small Hall	0.64	56	396	7.07	0.45	0.41	0.41	0.40	0.35	0.35	0.26
Gulbenkian Grand Hall	0.61	1203	7000	5.82	0.66	0.69	0.74	0.70	0.57	0.35	0.22
Lourinha Misericordia Church	0.47	50	1850	37.00	0.53	0.64	0.72	0.70	0.55	0.38	0.19
St Dominic Convent Church	0.48	–	6582	–	1.13	1.49	1.45	1.22	0.74	0.40	0.19
St George Cinema	0.53	830	3600	4.34	0.56	0.58	0.58	0.47	0.39	0.24	0.12
Newman	0.62	200	3350	–	1.63	1.30	1.50	1.55	1.28	0.34	0.19
Hormel	0.66	300	1710	–	1.44	1.13	0.96	0.97	0.94	0.34	0.20
Clarke	0.68	147	1300	–	0.90	0.71	0.66	0.66	0.61	0.34	0.19
Gusman	0.39	600	5,500	–	1.95	1.53	1.67	1.78	1.54	0.34	0.18

Table 2.3: Inverse Weighted Seven Octave Band Data

These three datasets converted into text files were read into MATLAB and used to develop the large room concert halls STI estimation equation. There also was an addition of a column called “Acoustic Density” which is defined by the following equation:

$$\text{Acoustic Density} = \text{Seat Capacity} / \text{Volume} \quad (2.5)$$

2.2.6 Model Architectures for STI estimation

Given the addition of the seven octave bands to the gathered data, a Multiple Linear Regression model (MLR), a Gaussian Process Regression model (GPR) and a Multi-Layer Perceptron model (MLP) were used to take in the seven octave band

reverberation features and make predictions on the STI. The MLP model had three hidden layers, one 64, the next 32, and the last was 16, there was a ReLu activation function applied to the output of each hidden layer. A 20% dropout rate for each of the first two hidden layers, and lastly a final output neuron for the STI prediction value. The MLP model was trained using the Adam optimization algorithm, with training performed for a maximum of 500 epochs and a batch size of eight. The training data were shuffled at the start of every epoch to promote generalization. As with any machine learning model, data must be separated to train and test the model. To ensure the robustness of the model, a pseudo random selection of halls was used to accomplish this task. Below is an example a train-test split of the halls.

Training Data	Testing Data
Cartuxa Church	Newman
Coliseu Micaelense	Gusman
Convent of Christ Church	Dom Joao III
Convent of Christ Cistern	Lourinha Misericordia Church
ESML Great Hall	
ESML Organ Room	
ESML Small Hall	
Gulbenkian Grand Hall	
St Dominic Convent Church	
St George Cinema	
Hormel	
Clarke	

Table 2.4: Sample Train-Test Split

After random selection, the models were trained on one subset of the data and tested on the other. Several iterations of each model were performed, and the resulting correlations, RMSEs, and MAEs were averaged to obtain the final evaluation metrics.

A GitHub repository with all recorded Room Impulse Responses (RIRs) captured during the on-campus measurements was created. This repository also includes a text file containing the extracted acoustic features for both the University of Miami halls and the 12 venues in the Reverbdata dataset. The repository can be accessed at: <https://github.com/NicolasAdler/UniversityOfMiamiConcertHallIRs>

2.3 Apparatus

Both the measurement and calculations performed in this thesis required many hardware and software tools. Brief descriptions of each tool are provided below.

2.3.1 Hardware Tools

- The Yamaha HS7 Studio Monitor was used as the source of the STI signal and the sine sweeps, it was picked because of its naturally flat frequency response.
- The Babyface Pro FS was used as the audio interface, it was picked for its clean response and therefore would leave minimal artifacts on the measured sound.
- G.R.A.S 26CA Microphone is a reference microphone with a low noise floor and frequency response from below 2 Hz up to 200 kHz.

- XLR cables, one which was 5 feet long, the other which was 50 feet long. The 5 foot XLR cable was used to connect the output of the Babyface Pro FS to the Yamaha HS7, the 50 foot XLR was used to connect the reference microphone to the input of the Babyface Pro FS.
- Mac Book Pro M3 to collect the data, play out the test signals, EQ the HS7 after its frequency response test, and to run Room EQ Wizard (REW).
- Tape Measure to measure the distance from the driver to the SPL meter to ensure that the test signal played out of the HS7 was playing at 65 dB_{SPL} for the STIPA signal, and 80 dB_{SPL} for the sine sweeps.
- Laser Measure to measure the heights at critical location in each of the concert halls at the University of Miami. Only floor views of the concert halls were provided, so manual height measurement was required to calculate the volumes of the concert halls.
- Larson & Davis XL8 SPL meter used alongside the G.R.A.S 26CA to calibrate REW for the room measurements.

2.3.2 Software Tools

- Room EQ Wizard (REW). A powerful free software that allows users to easily view acoustic data. REW can create audio interface calibration files, and allows importing of microphone calibration files as well. If needed, the output of sweeps could be equalized to reduce any artifacts imprinted on the sound by the source speaker.

- MATLAB. Used to develop and run the scripts for calculating the RT60s and the STI measurements of the 12 different venues in the Reverbdata dataset, calculate p -values, and create the Multiple Linear Regression and the Gaussian Process Regression models.
- Microsoft Excel. Used to store the measurements, create charts and perform basic analysis.
- EasyPDFTakeOff. This is a free software similar to Bluebeam. EasyPDFTakeOff allows for users to measure distances in PDFs using a scale size such as 1/8 in = 1 ft. All the blueprints for the concert halls on campus had this 1/8 in to 1 ft scale; feet to meter conversion were used to keep consistency with the metric calculations of the halls in the Reverbdata set.
- Logic Pro. Once all of the STI and Sine Sweep measurements were completed in each hall, wooden clap RIRs were recorded into Logic Pro.
- StudioSixDigital STI/STIPA application. This app was used for the measurements of STI in each of the concert halls on campus. The application can generate the warble tone used in standard STI measurements, and can take in microphone calibration files. After about 15 seconds of “listening” to the warble tone, the application will report the STI measurement.

CHAPTER 3

Results

Prior to discussing the results between the original five equation's predicted STI and calculated STI, it is important to investigate the quality of the system used to produce the RT60 values. A paired t -test was used comparing the reported RT values by the authors of the data set and the calculated RT values from the scripts, the p -values for this test was $p=0.967912$.

3.1 Simulation Phase

The following section will be used to describe the initial results from the simulation phase of the experiment. These results are entirely based off of the Reverbdata set, and scripts written that would extract the necessary features for the calculations.

Each RIR wav file was passed into the RT60 at 500 Hz script to produce the RT values at that exact location in the venue. Those RT60 values were then passed into the STI script to get a calculated STI value for the exact same position. For each equation, at each venue a chart was created with the calculated STI, and the predicted STI. Below is one of these charts; it displays equation two's calculated and predicted STIs in St. Dominic Convent Church.

Equation 2 — St. Dominic Church Calculated vs Predicted STI Values

Calculated STI	Predicted STI
0.603043	0.407494
0.604476	0.398901
0.495965	0.418095
0.464088	0.400744
0.410889	0.414341
0.455695	0.404533
0.461236	0.405627
0.395152	0.412586
0.434302	0.408601
0.603041	0.407494
0.604471	0.398901
0.495968	0.418095
0.464055	0.400744
0.410901	0.414341
0.455699	0.404533
0.461248	0.405627
0.395175	0.412586
0.434334	0.408601
0.603041	0.407494
0.604471	0.398901
0.495968	0.418095
0.464055	0.400744
0.410901	0.414341
0.455699	0.404533
0.461248	0.405627
0.395175	0.412586
0.434334	0.408601

Table 3.1: Calculated Vs. Predicted STI Data

A resultant chart was created with the p-values for each equation at each venue.

Equation 1	Cartuxa Church	Coliseu Micaelense	Covent Church	Covent Cistern	Dom Joao III	ESML Great Hall	ESML Organ Room	ESML Small Hall	Gutbenkian Hall	Lourinha Church	St Dominic Church	St George Cinema
T-Test P val	0.00015159	0.00085456	0.00002252	0.00070211	0.04577812	0.00000000	0.00092850	0.00791957	0.00006102	0.00011271	0.00000002	0.36299129
Corr R val	0.31410524	0.01300420	0.26148130	-0.42443498	0.29027930	-0.55172164	-0.07583449	0.26603817	0.81332514	-0.47671151	-0.42989880	0.47433139
Sig of Cor t val	1.75069458	0.05036931	1.08362615	-1.48232236	0.95924702	-2.56203378	-0.32266766	0.78060047	8.61737992	-2.30077137	-2.38071608	1.78700034
Sig of Cor p val	0.09094958	0.96049271	0.29459061	0.16906711	0.36005243	0.02167440	0.75066482	0.45751073	0.00000000	0.03357568	0.02521510	0.10149157
Equation 2												
T-Test P val	0.00000009	0.04161628	0.00000001	0.00000000	0.00312436	0.00000000	0.00000001	0.58944113	0.00882308	0.00000000	0.00000000	0.00007729
Corr R val	0.31178429	0.01412715	0.26330924	-0.42498635	0.29071007	-0.53165278	-0.07634254	0.27217678	0.80342480	-0.48177840	-0.43471018	0.47057124
Sig of Cor t val	1.73635986	0.05471966	1.09176366	-1.48467213	0.96080185	-2.41137383	-0.32484198	0.80003572	8.31809062	-2.33256736	-2.41352687	1.68646985
Sig of Cor p val	0.09349294	0.95708398	0.29110474	0.16845476	0.35930736	0.02806118	0.74904570	0.44679380	0.00000000	0.03147815	0.02345468	0.12260128
Equation 3												
T-Test P val	0.00000000	0.76964574	0.00000012	0.69346862	0.50943712	0.00000000	0.00000020	0.40424586	0.00661634	0.00000000	0.00003509	0.00053707
Corr R val	0.31412545	0.01299980	0.26153105	-0.42449078	0.29025786	-0.55172135	-0.07587907	0.26603559	0.81048102	-0.47669225	-0.42989806	0.49042987
Sig of Cor t val	1.75081950	0.05035227	1.08384743	-1.48256008	0.95916967	-2.56203185	-0.32285840	0.78059232	8.52925915	-2.30065112	-2.38062278	1.77958671
Sig of Cor p val	0.09092767	0.96050606	0.29449542	0.16900507	0.36008953	0.02167448	0.75052274	0.45751527	0.00000000	0.03358386	0.03395028	0.10549782
Equation 4												
T-Test P val	0.00000000	0.89516685	0.00000043	0.48702194	0.43614250	0.00000000	0.00000000	0.05342905	0.01237992	0.00000000	0.00001844	0.00004900
Corr R val	0.31410699	0.01299741	0.26153979	-0.42438501	0.29024165	-0.55172127	-0.07578536	0.26603833	0.81047724	-0.47670966	-0.42987891	0.47433044
Sig of Cor t val	1.75070537	0.05034300	1.08388631	-1.48210952	0.95911117	-2.56203129	-0.32245741	0.78060097	8.52914300	-2.30075986	-2.38058101	1.78699572
Sig of Cor p val	0.09094768	0.96051332	0.29447870	0.16912267	0.36011759	0.02167450	0.75082145	0.45751046	0.00000000	0.03357647	0.02522259	0.10149235
Equation 5												
T-Test P val	0.00000092	0.00749856	0.00000308	0.05584527	0.14525432	0.00000000	0.09226270	0.02474506	0.00015591	0.00000294	0.00000020	0.13665171
Corr R val	0.31410524	0.01300420	0.26148130	-0.42443498	0.29027930	-0.55172164	-0.07583449	0.26603817	0.81047902	-0.47671151	-0.42989880	0.47433139
Sig of Cor t val	1.75069458	0.05036931	1.08362615	-1.48232236	0.95924702	-2.56203378	-0.32266766	0.78060047	8.52919773	-2.30077137	-2.38071608	1.78700034
Sig of Cor p val	0.09094958	0.96049271	0.29459061	0.16906711	0.36005243	0.02167440	0.75066482	0.45751073	0.00000000	0.03357568	0.02521510	0.10149157
Equation AVG												
T-Test P val	0.00000001	0.06772016	0.00177076	0.00003025	0.10419698	0.00000000	0.00025432	0.69722585	0.00157486	0.00000000	0.00000011	0.00197658
Corr R val	0.31340679	0.01333758	0.26234996	-0.42470317	0.29044375	-0.54260804	-0.07594293	0.26697885	0.80893143	-0.47787929	-0.43171155	0.47364936
Sig of Cor t val	1.74637711	0.05166081	1.08749165	-1.48346502	0.95984052	-2.50184152	-0.32313171	0.78357201	8.48199157	-2.30807291	-2.39304848	1.78368697
Sig of Cor p val	0.09170937	0.95948066	0.29293094	0.16876909	0.35976790	0.02441689	0.75031916	0.45586106	0.00000000	0.03308285	0.02453966	0.10205298

Table 3.2: All statistical values for Reverbdata STI predictions.

Table 3.2 displays all the statistics conducted on all the equations for each of the halls in the Reverbdata dataset, the p -values that were derived from a paired t -test which did not produce a statistically significant difference are in light green cells. The values in the dark green cells indicate p -values that are greater than $p = 0.05$, which indicate a strong correlation between the calculated and predicted STI values.

A two-way analysis of variance (ANOVA) was performed to see the effect of volume, RT60 at 500 Hz, and these features working together on the STI metric.

The results of the two way ANOVA are presented below:

Source	Sum Sq.	d.f.	Singular	Mean Sq.	F	Prob > F
Volume	0.00787	1	0	0.00787	0.578	0.4632
RT60	0.04201	1	0	0.04201	3.085	0.1068
Volume:RT60	0.00896	1	0	0.00896	0.658	0.4345
Error	0.14981	11	0	0.01362		
Total	0.22256	14	0			

Table 3.3: Two-Way ANOVA of Volume, RT60, and Interaction Effects on STI

While none of the values from this ANOVA reveal a feature that was significant, the p -values for both the volume in isolation and the interaction between volume and RT60 show that volume in this data, was not a good predictor of STI.

3.2 Measurement Results

Once the simulation phase was complete, the measurements were gathered in the four concert halls on campus. Just as was done in the simulations phase, each measurement position had four sine sweeps with four respective STI measurements, the following chart shows the average RT60s at 500 Hz and STI for each of the concert halls.

Halls	Newman	Hormel	Clarke	Gusman
RT60 at 500 Hz (s)	1.68	1.08	0.74	1.88
STI	0.62	0.66	0.67	0.39

Table 3.4: RT60 at 500 Hz and STI Measurements for Each Hall

Similarly to what was done on the simulation phase data, the RT60 values were passed into each of the original five prediction equations. Several statistical tests were used to find the significance of the predictive power of the equations. Below are the results.

Equation1	Newman	Hormel	Clarke	Gusman
T-Test P val	0.00000000	0.00000000	0.00000000	0.00000113
Corr Rval	0.71051805	0.07164892	0.28373551	0.16035926
Sig of Cort val	5.53045889	0.39344851	1.62068955	0.88983951
Sig of Cor p val	0.00000521	0.69677062	0.11554956	0.38063254
Equation 2				
T-Test P val	0.00000000	0.00000000	0.00000000	0.00000000
Corr Rval	0.70524718	0.07164892	0.28147158	0.15890146
Sig of Cort val	5.44852983	0.39344851	1.60664054	0.88153956
Sig of Cor p val	0.00000657	0.69677062	0.11860921	0.38503444
Equation 3				
T-Test P val	0.00000000	0.00000000	0.00000000	0.21848974
Corr Rval	0.71051805	0.07164892	0.28373551	0.16035926
Sig of Cort val	5.53045889	0.39344851	1.62068955	0.88983951
Sig of Cor p val	0.00000521	0.69677062	0.11554956	0.38063254
Equation 4				
T-Test P val	0.00000000	0.00000000	0.00000000	0.06552618
Corr Rval	0.71051805	0.07164892	0.28373551	0.16035926
Sig of Cort val	5.53045889	0.39344851	1.62068955	0.88983951
Sig of Cor p val	0.00000521	0.69677062	0.11554956	0.38063254
Equation 5				
T-Test P val	0.00000000	0.00000000	0.00000000	0.00019594
Corr Rval	0.71051805	0.07164892	0.28373551	0.16035926
Sig of Cort val	5.53045889	0.39344851	1.62068955	0.88983951
Sig of Cor p val	0.00000521	0.69677062	0.11554956	0.38063254

Table 3.5: All p -values for Campus STI predictions.

The p -values for both the paired t -test and the Pearson's correlation significance p -values display are a more extreme version of the results that were shown in the simulations phase. Other than two exceptions (one of which is very close to the α threshold of $p=0.05$, the vast majority of the paired t -test p -values in this chart show that $p < 0.01$. Likewise the vast majority of the correlation p -values are greater than 0.05 meaning low correlation between the measured and predicted STI values.

The results from both the simulation phase and the on-campus measurements support the original hypothesis; the existing STI estimation equations, which were derived largely from primary school classrooms, are not able to accurately predict the speech transmission index for large more reflective spaces. Averaging the equations likewise never indicated any form of improvement in the predictive ability of the equations.

3.3 New Models

The following section reports the results from the various models that were trained on the Reverbdata and University of Miami concert halls.

3.3.1 Model 1: All collected data into the Multiple Linear Regression Model

All three new datasets, the original seven octave band data followed by the weighted and inverse weighted octave band data was fed into an MLR model. The R^2 values ranged between 0.819 to 0.974, where the weighted STI data received an R^2 value of 0.819, the original data received an R^2 value of 0.961 and the inverse weighted data received an R^2 value of 0.974. The original data had the lowest RMSE of the three, with an RMSE of 0.0414. The mean absolute error for model 1 was 0.2127 STI units.

3.3.2 Model 2: Picking the most important octave bands

Model 2 used a subset of the total T60 data, the subset excluded the outer bands, 125 Hz and 8000 Hz. Model 2's performance was worse than Model 1, which was also poor. Model 1 had a very high R^2 value of 0.961, but was overfitting, indicated by the high MAE value of 0.2127. While Model 2 had a lower R^2 value, 0.742, with a RMSE of 0.068, its MAE was 0.2164.

After several iterations of the outer octave band removal (as in 250 to 4000 Hz, then 500 to 2000 Hz), the best results came from using the 250 Hz to 4000 Hz octave bands.

This model had an R^2 value of 0.42, and RMSE of 0.1463, and a much lower MAE of 0.1404. The resulting equation is listed below:

$$STI = 0.6934 + 0.0426 \cdot T60_{250} - 0.1839 \cdot T60_{500} + 0.1954 \cdot T60_{1000} - 0.5787 \cdot T60_{2000} + 0.4767 \cdot T60_{4000} \quad (3.1)$$

This model was subjected to the same statistical tests that were used in the simulation phase and measurement phase data. Table 3.1 displays these results for 10 iterations of the MLR model. The same trend occurred with this equation as with the others decried before, where no paired t -test p -values were able to produce also produce significance correlation values. Model 2's average MAE after the 10 iteration was 0.11519, a slight improvement over model 1's MAE of 0.1404.

MLR	Run 1	Run 2	Run 3	Run 4	Run 5	Run 6	Run 7	Run 8	Run 9	Run 10
t-test p val	0.00754	0.00289	0.01258	0.12032	0.01248	0.57164	0.00903	0.39822	0.00323	0.01371
corr r val	0.4503	0.2646	0.0685	0.5807	0.5569	0.1195	0.6988	0.6001	0.3864	0.8712
Sig of Corr t val	1.887	1.0267	0.2568	2.6686	2.509	0.4503	3.6556	2.8071	1.5675	6.6385
Sig of Corr p val	0.08007	0.322	0.80107	0.01835	0.04569	0.65938	0.0026	0.01398	0.13931	0.00001
RMSE	0.1463	0.1586	0.1647	0.1683	0.0963	0.2021	0.1764	0.1285	0.1495	0.0966
MAE	0.1144	0.1091	0.1357	0.1357	0.0673	0.1736	0.1278	0.1021	0.1219	0.0643

Figure 3.1: Model 2, All stats

3.3.3 Model 3: Lasso-selected Model

A similar approach of truncating the octave bands was used on the original data, but with the lasso technique. After several iterations the same effect as seen in Model 2, where the central octave bands were producing better results, was also seen here. Single octave bands were then used to train the model, with T60 at 500 Hz producing the best and simplest model thus far.

$$STI = 0.6117 - 0.0833 \cdot T60_{500} \quad (3.2)$$

This equation had an RMSE of 0.1495 and an MAE of 0.1428. This model performs slightly worse than Model 2 but has a more simple equation with only one parameter needed, the RT60 at 500 Hz.

3.3.4 Model 4: Gaussian Process Regression

A similar approach was used to achieve marginal improvements on output of the GPR, where the used bands were altered across different iterations to find the exact bands that would produce the lowest error in testing. For the GPR the best performance was using the 250 and 500 Hz bands. The output of the code is essentially a MATLAB function that can be inferred from. The GPR's Model Test RMSE was 0.1675, and its Test MAE was 0.1531. Below is the output of 10 iteration of the GPR model.

GPR	Run 1	Run 2	Run 3	Run 4	Run 5	Run 6	Run 7	Run 8	Run 9	Run 10
t-test p val	0.00421	0.00009	0.39625	0.01734	0.01746	0.25309	0.32276	0.00795	0.20473	0.86339
corr r val	0.6477	0.5356	0.1019	0.6999	0.3912	0.251	0.5158	0.4465	0.7329	0.3825
Sig of Corr t val	3.1812	2.3733	0.3832	3.6663	1.5907	0.9701	2.2527	1.8673	4.0312	1.5488
Sig of Corr p val	0.00667	0.03249	0.70733	0.00254	0.13401	0.34846	0.04085	0.08293	0.00124	0.14372
RMSE	0.2023	0.1917	0.2328	0.1468	0.1352	0.1263	0.1444	0.1408	0.1659	0.1911
MAE	0.1462	0.159	0.2053	0.1061	0.1084	0.0914	0.1218	0.1225	0.1456	0.1602

Table 3.6: Table of GPR Results

3.3.5 Model 5: Multi-Layer Perceptron

As with the previous two architectures, a similar approach of truncating the full seven octave band data was used in order to achieve the lowest error for the MLP. For the MLP the best performance on the given data was found by using the 500 to 2000 Hz octave bands. Similar to the GPR, after training the MLP, the output from the training is a function that a user infer from. After 10 iterations of this model the average RMSE was 0.14283 and the average MAE was 0.10629.

MLP	Run 1	Run 2	Run 3	Run 4	Run 5	Run 6	Run 7	Run 8	Run 9	Run 10
t-test p-value	0.0149	0.1258	0.0655	0.9507	0.0235	0.7844	0.748	0.0377	0.7156	0.4235
R Value	0.4774	-0.0304	0.1739	0.3523	0.7223	0.8293	0.7355	0.5163	-0.4465	-0.1799
t-statistic	-2.7495	-1.6213	-1.9868	-0.0628	-2.5209	-0.2785	-0.3273	-2.279	0.3713	0.8228
correlation p-value	0.0615	0.911	0.5196	0.1807	0.0016	0.0001	0.0012	0.0406	0.0829	0.5049
RMSE	0.1759	0.2052	0.1475	0.0828	0.0904	0.0939	0.0866	0.1729	0.1947	0.1784
MAE	0.1173	0.1574	0.096	0.065	0.0716	0.0718	0.0675	0.1137	0.1607	0.1419

Table 3.7: Table of MLP Results

3.3.6 All Model Results

The chart below combines all the results from the five equations on measurement phase data, and the results from the MLR, GPR, and MLP models. Tabel 3.8 indicates in bold the lowest error values across the existing equations and new models for both RMSE and MAE.

	Equation 1	Equation 2	Equation 3	Equation 4	Equation 5	GPR	MLR	MLP
RMSE	0.1841	0.2001	0.1462	0.1449	0.1726	0.1677	0.1487	0.14283
MAE	0.1749	0.1917	0.1316	0.1347	0.1636	0.1366	0.1159	0.10629

Table 3.8: RMSE and MAE results for all models and existing STI equations.

CHAPTER 4

Discussion

This section will investigate the results from the simulation phase, measurement phase, and the new model prediction accuracy.

4.1 Simulation Phase

This simulation utilized the Reverbdata dataset, created for the 2024 European AES convention. It contains wav files of room impulse responses from 12 medium-large halls in Portugal. The choice of this dataset is that it is a dataset that matches the definition given in [4], which states that medium-large halls are considered to be those “[whose] volume [is] higher than 350 cubic meters”. All halls in this dataset are larger than 350 cubic meters and are not classrooms.

Given that in [4], the authors mentioned equation 4.5 was the most accurate for medium and large rooms, and that 4.6 performed the worst for the small classroom setting it may have been unnecessary to test all six equations. However when considering the measurements that were used to derive these two equations: equation 4.5 only using primary school classrooms, and equation 4.6 being the only equation in [4] that was developed using larger reflective spaces, it was safest to test all the equations in the new condition, rather than disregarding any due to results that were unrelated to this current work.

Table 3.2 displays the results for the five existing equations on the 12 venues in the Reverbdata dataset. As mentioned earlier, the light green cells indicate paired t -test p -values that were above the α threshold of $\alpha = 0.05$, this indicates prediction STI values that are close in value to the calculated STI values. The darker green cells display p -values derived from Pearson’s correlation test, where values that were below the α threshold of $\alpha = 0.05$ indicate a significant correlation between the calculated and predicted STI values. A significant correlation means that as the calculated STI metrics drop, the predictions also drop, and vice versa. A non-significant paired t -test p -value (light green cell) along with a significant correlation p -value (dark green cell) would indicate an equation that has a close prediction accuracy that also follows the trend of the calculated STI metrics and would likely indicate a good model. At no point in Table 3.2 is there an equation or venue that could produce a light green cell and dark green cell. By simply counting the total non-significant paired t -test p -values for each equation reveals that no equation could achieve a non-significant difference between the calculated and predicted STI values more than a third of the time. The same can be said about the correlation values. Thus no existing model can yield an accurate STI metric for the large and more reflective venues in the Reverbdata dataset. The data in this table is valuable in that it provides more evidence toward the validity of these existing equations for smaller spaces. The appropriately named ESML Small Hall, indeed was the smallest hall in this dataset, with a volume of 396 cubic meters; while that volume is considered to be a medium-large hall according to [4], three of the equations were able to produce p -values that had no statistically significant difference between the predicted and calculated STI values. The Convent of Christ Cistern, produced a low yield of statistically significant p -values as well. The cistern’s volume is also low when compared to the rest of the data, its volume is 767 cubic meters. The Dom Joao III church provides some evidence in favor of the predictive power of the equations; the Dom Joao III church had the smallest number of significant p -values, a total of four out of the five

equations were able to predict STI correctly. Yet despite these details, the overwhelming majority of the statistical values point towards the need for improvement on the predictive accuracy of the existing models in non-classroom spaces.

4.2 Measurement Phase

The results displayed in Table 3.5 are a stronger version of the results in the simulation phase data shown in Table 3.2; the overwhelming majority of the results show that no equation could produce non-significant paired t -test p -values. There was a total of only two light green cells in this chart, with one of two of them being very close to the α threshold of $\alpha = 0.05$. Besides Newman recital hall, which achieved a significant correlation for all the equations, no other halls could achieve this metric. Ultimately the results from both the simulation phase data and the campus measurement data support the alternative hypothesis of this thesis, the existing equations are not able to accurately predict the STI in larger more reflective non-classroom spaces.

4.2.1 Existing Models

As shown in [4–7] all six equations (models) are either linear equation models or logarithmic equation models; the choice of these two models types are not arbitrary, rather they logically makes sense. It was expected to see an inverse relationship between the volume of spaces and the STI. While none of the estimation equations found in [4–7] directly have a volume coefficient, this parameter is imprinted into the RT60 values. The volume of a space has a large impact on the reverberation time, thus, in most cases RT60 times and STI index values have an inverse relationship. When using data with only a few features, primarily RT60 at 500 Hz, as is the case with both the original estimation equations and those developed in this project, the resulting models tend to be either linear or logarithmic

in form. However, the results from the ANOVA that was performed to see the effect and interaction of Volume and RT60 on the STI metrics were surprising. To fully understand the result from this ANOVA its best to set up some hypotheses. From what is generally known about STI in room acoustics, the alternative hypothesis would state: “If the volume of a room is changed, which would simultaneously change the RT60 of that space, the STI of that space would also change.” The null hypothesis would then state: “Changes in volume or RT60 do not result in changes in STI.” The ANOVA reveals that none of these features, nor their combination, were able to produce a significant p -value. The lowest p -value indeed did come from the RT60 feature, and had the α threshold been more lenient, such as an $\alpha = 0.1$, and given more data, the RT60 feature would likely have trended toward significance. Another indication of this phenomenon was seen with the during the development of the MLR models. Whenever the model was presented volume as a input feature to train the model, the output equation would have a coefficient for the volume feature that was close to 0.00 or simply just 0.00. This suggests that the volume data, for this given dataset was not a good feature at accurately predicting the STI metric. This finding was indeed unexpected, however, it was supported by the calculated and measured data. Table 2.1 displays the original data with averaged RT60 values for all seven octave bands. This table shows several halls, like Newman recital hall, the Gulbenkian Hall, and Hormel that each have large volumes, well above the 350 cubic meter threshold; yet their STI metrics are high, each falling into the “Good” category as shown in figure 1.1.

Upon further investigation, it can be inferred from the data that halls that serve as multipurpose spaces tend to result in poorer STI predictions and could be the reason for the described phenomenon above. For example the Gulbenkian hall, which “with [its] exceptional acoustics, design and construction [was] aimed at hosting many cultural activities, including symphonic and choral-symphonic music, congresses and conferences, ballet, theater, and cinema” [21]. The same applies to the ESML Great Hall, which “allows a variety

of cultural events such as theater, cinema, concerts, and conferences.” The act of acoustically treating these spaces might be the reason the prediction of STI is inaccurate. There were many of these types of halls in the measured data as well, for instance, Hormel is a space used for recordings, performances, event gatherings, and music festivals such as the yearly CAM fest at the University of Miami. Newman is used for recitals and doctoral lecture presentations, Clarke is often used for contemporary music performances, classical chamber showcases, and lectures as well. Newman for instance has high ceilings, resulting in a large volume, narrow walls, and a rather short distance from the speaker to the audience, Newman likely has a high C50. A large chunk of the venue data fed into the new models contains these multipurpose spaces that have large volumes and reasonably good STI. Equations 4.5 to 4.9 were all mostly derived via measuring small classrooms, and some equations entirely derived from primary school classrooms alone. While not always the case, classrooms such as these are usually never subjected to such extreme acoustic treatment, largely because their size deems this level of treatment unnecessary. Large halls are often the byproduct of extraordinary engineering; this engineering clearly manipulates the sound in such ways that it makes what traditionally would be thought to be impossible, possible. A hall as large as the Gulbenkian Grand Hall having an STI of 0.64, considered to be “Good” is not an expected result, and yet this hall does not stand alone as the only surprising result, Newman, The ESML Great Hall, and Hormel are also great examples. If the original measurement data did not take into account larger spaces with acoustic treatment, it becomes clear as to why the prediction of STI is under-performing in these spaces.

On the contrary, some spaces due to their historic background have very little treatment at all, and by checking their RT60 values reveal that these spaces are highly reflective despite having a small volume. One such example is the Convent of Christ Cistern, one of the smallest venues in the Reverbdata dataset, yet is tied with the ESML Great hall for the longest RT60 at 5.12 seconds. The image of this space found in [21] depicts a room

made entirely of stone and with absolutely no acoustic treatment. When considering this explanation it can be reasonably inferred why a model given this particular dataset would find that the volume feature is not the best for STI prediction.

4.2.2 New Models

In the methods section of this thesis are three different tables that plot all the data collected, those were Tables 2.1, 2.2, and 2.3. The first thing to notice about this data is the addition of the full seven octave band data; the addition of the other octave bands came from [7], where the authors achieved a significant improvement for the STI prediction, resulting in an $R^2 = 0.971$. They attributed the high prediction accuracy to the inclusion of neighboring octave bands: 125 Hz and 1000 Hz, as opposed to using only 500 Hz. Since all of this T60 data was easily accessible by simply changing the script that was used to create the RT60 data to include all the frequencies needed to calculate the STI, all the seven octave bands present in the STI calculation were used for the creation of the models. Another important thing to note about these charts was the application of the weighting factors determined by [14], to the second chart, and the inverse of these weighting factors to the third chart. The idea behind this was a simple pre-processing step for the model, since STI is calculated using these weights but the calculated seven octave band data originally did not have these weights applied, the weights might have helped out the model increase its prediction accuracy by “filtering” the RT60 into the format that STI is calculated with. The results from the MLR Model 1 revealed that the addition of the weights did the opposite effect. The inverse weights in fact had the highest R^2 value of 0.974, while only the weights had the lowest ($R^2 = 0.819$). The difference between the inverse weight and the original data was marginal, where the inverse R^2 was 0.961 and the original R^2 was 0.974. For the sake of simplicity of the model, all the models thereafter were trained using the original data. The choice of the MLR architecture for the creation of a new model stems from the

addition of the seven octave band data. Multiple Linear Regression models are able to take in two or more features and fit a line to predict the target. In this thesis the target of course is the STI index, and the features were the T60 values for all the octave bands (or a subset of these bands), volume, and the acoustic density. To set up the models, 10 of the 12 Reverbdata dataset halls along with two of the four halls on campus were used to train the model. While traditional approaches would choose these data points entirely randomly, a pseudo random approach was used for this experiment. This was to ensure that not all of the testing or training data contained only simulation phase data, as it would be better to measure the quality of the data on the real measurements. As stated earlier, when comparing the results of the existing equations on the Reverbdata compared to the measured data, the measured data revealed a more extreme version of the simulation results. While Table 3.2 showed that the existing equations could not accurately predict STI in these larger more reflective spaces, Table 3.5 was an even more extreme version of Table 3.2. Thus if new models could be tested against real measurements and achieve a high prediction accuracy, the new models would be technically robust.

4.2.3 MLR GPR, and MLP VS. The Existing Equations

Both Table 3.1 and Table 3.6 did see a marginal improvement in statistical test accuracy when compared to the existing equations, however this improvement is quite nuanced. For example, in Table 3.2, it can be seen that at best the existing models could produce non-significant paired t -test p -values was a third of the time, or 33%, the same could be stated about the correlation values. MLR Model 2, and the GPR after 10 iterations both produced significant correlations and non-significant paired t -test p -values for five out of the 10 iterations, or 50%. However, when looking at the final error of both of these new models, their improvements are too marginal to deem any meaningful improvement to the existing models. While the statistical values show the trends of the predictions compared

to the calculated STI values, what an accurate and robust model needs is low error. Table 3.8 displays the RMSE and the MAE of all the five existing equation and the MLR, GPR, and MLP models side by side. The model with the lowest RMSE and MAE was the Model 5, the Multi-Layer Perceptron. It was the only model that was able to achieve a MAE that was near 0.1 STI units. The RMSE for this model was 0.14283 and the MAE for this model was 0.10629, very close to 0.1 STI units. Ultimately this error is still too high to be considered useful for real world application; an error this large could give a venue a “Good” intelligibility category, as shown in Figure 1.1, despite its true rating being closer to “Poor” or “Fair.” However, when compared to the existing equations, Model 5 is a clear improvement to the existing models. The MLP was the only model in the entire thesis that was able to achieve a non-significant paired t -test p -value and a significant correlation p -value. This can be seen in Table 3.7 under Run 6 and Run 7. Where both the non-significant paired t -test p -values and significant correlation p -values were displayed in red text. Given more data, especially more specific data, i.e. only concert halls, the MLP might be able to produce STI predictions that consistently have MAE errors around 0.05 STI units or less.

4.3 Limitations

Although some of the new models had a higher prediction accuracy than the existing equations, they ultimately are still not accurate enough to provide any meaningful metric. This is largely due to the dataset sample size, while the Reverbdata dataset has over 200 RIR recordings, there were only 12 halls measured in the data. Likewise, only four halls were added on top of this Reverbdata set. Performing a standard train test split to the data meant that roughly 12 halls were used to train, and four halls were used to test. To get a strong model that will not overfit this small dataset is an extremely difficult task, using different models such as the GPR, or the use of techniques like the Lasso technique can

only increase the accuracy of the equations marginally and ultimately did not make notable increases in accuracy. Besides the small dataset, a few other factors could be leading to lower accuracy. While [22], the creators of the Studio Six Digital application, state that their tool “results in an STI estimation that is fully compliant with [23]”, it still is not the traditional method of measuring STI. For the most accurate results it would have been best to use a real STI meter and Talkbox.

The Reverbdata also contained many historic churches as data, and while meeting the criterion of a large reflective space, they also posed other problems. The most notable was the lack of seat count data for all of the spaces. Even though only a few halls were missing the seat count information, this further truncated the already small dataset, making one of the most prominent limitations in this thesis even more pronounced. To combat the truncation of the full 16 hall dataset due to missing information, the entire acoustic density feature was removed during the model creation phase; this is less than ideal as it could have been a significant feature to predict STI especially alongside the volume feature.

One of the largest limitations to both the new models and the existing equations comes from the lack of consideration in the equations for background noise. A concert hall, classroom, or common space may have been engineered to have “Excellent” STI by using an improved version of the MLP model or the existing equations, but neither of these estimation methods takes into account background noise or the signal-to-noise ratio of the space. If a loud projector is placed in this “perfectly” engineered space, it would most certainly affect the STI. The same applies to an HVAC system, or outside construction noise entering the acoustic environment. Without this consideration, no model would truly have a high accuracy, regardless of what any statistical analysis says.

4.4 Future Work

4.4.1 Data Collection

As discussed earlier, the biggest drawback to this experiment was the limited dataset. Acoustics is already a niche field, and STI is a niche within that niche. At the time of writing, there are practically no publicly available STI datasets beyond what has been measured or calculated in this thesis. Expanding this dataset would be one of the most impactful ways of improving estimation models. Future studies should prioritize collecting STI data from a variety of environments, particularly from spaces where speech intelligibility is known to be critical. These include food courts, airports, restaurants, lecture halls, churches, and concert venues. More data across a broader range of environments would open the door to more robust statistical analyses, better feature selection, and better estimation models.

4.4.2 Parallels to other Intelligibility metrics

In the case of concert halls, when a hall requires better speech intelligibility but has poor intelligibility metric the simplest solution is amplification. Research into predicting STIPA in a hall given a specific acoustic system would likewise be a very useful metric. A tool that could import a frequency response datasheet of a speaker, incorporate some STI estimation, then produce a STIPA estimation would be the fully realized version of this whole thesis.

4.4.3 SNR or Background noise incorporation

As discussed earlier, a room that may be near a highly noisy space, or have a noisy item placed in it would be susceptible to having an inaccurate STI estimation, especially when

none of the models incorporate an element of SNR. A future model which could take into account this detail would be the most robust model.

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APPENDIX

Key Equations

Reverberation Time (Sabine Equation)

$$\text{RT60} = 0.049 \times \frac{V}{A} \quad (4.1)$$

Total Absorption

$$A = \sum_i S_i \cdot \alpha_i \quad (4.2)$$

Clarity Index (C50)

$$C_{50} = 10 \log_{10} \left(\frac{D_{50}}{1 - D_{50}} \right) \times 100\% \quad (4.3)$$

Definition Index (D50)

$$D_{50} = \frac{E_{\text{early}}}{E_{\text{total}}} \quad (4.4)$$

STI Estimation Equations

$$\text{STI}_1 = 0.5895 - 0.4422 \log_{10} (RT60_{500 \text{ Hz}}) \quad (4.5)$$

$$\text{STI}_2 = 0.778 - 0.143 (RT60_{500 \text{ Hz}}) \quad (4.6)$$

$$\text{STI}_3 = 0.634 - 0.192 \ln (RT60_{500 \text{ Hz}}) \quad (4.7)$$

$$\text{STI}_4 = 0.6488 - 0.2078 \ln (RT60_{500 \text{ Hz}}) \quad (4.8)$$

$$\text{STI}_5 = 0.5940 - 0.1827 \ln (RT60_{500 \text{ Hz}}) \quad (4.9)$$

Average STI Estimation

$$\text{STI}_{\text{avg}} = \frac{\text{STI}_1 + \text{STI}_2 + \text{STI}_3 + \text{STI}_4 + \text{STI}_5}{5}$$